Zero-cross Detection

0.05Vrms~1Vrms
30Hz~150kHz

J1
B2B-XH-A(LF)(SN)

C3
1u PET

U2A
OPA2140AIDGKR

R5
4.7k

10M R2

3k R3

100 R6

4.7k R8

C11
0.1u

C12
0.1u

LM311
U4

1k R9

REFERENCE_SIGNAL

Zero-cross Detection
Offset Adjustment

R4
50k

U2B
OPA2140AIDGKR

U2C
OPA2140AIDGKR

C18
0.1u

C19
0.1u

C45
1.5u

SW1

100k R18

D2
1N4148

RESET

Y1
AE-SIT2001B-12MHz

C27
0.1u

47 R10

CLK_12MHz

U5B
10M02SCE144C8G
BANK2
IO/CLK0n
IO/CLK0p
IO/CLK1n
IO/CLK1p
IO/VREFB2N0
IO/PLL_L_CLKOUTn
IO/PLL_L_CLKOUTp

U5E
10M02SCE144C8G
BANK6
IO/CLK2p
IO/CLK2n
IO/CLK3p
IO/CLK3n
IO/DPCLK3
IO/VREFB6N0
IO/DPCLK2

U5A
10M02SCE144C8G
BANK1
IO/JTAGEN
IO/VREFB1N0
IO/TMS
IO/TCK
IO/TDI
IO/TDO

U5C
10M02SCE144C8G
BANK3
IO/VREFB3N0

SQ_OUT_90
SQ_OUT_0
LOCKED

J3
B3B-XH-A(LF)(SN)

J4
B2B-XH-A(LF)(SN)

J5
Conn_01x10

TCK
TDO
TMS
TDI

10k R13
10k R14
10k R11

U3
TJ7660N

5V

J2
B2B-XH-A(LF)(SN)

C1
33u

C5
10u

C6
0.1u

C9
10u

C10
10u

0 R1

U1
AZ1117CH-3.3TRG1

C2
33u

C4
0.1u

C7
0.1u

C8
33u

4.7k R7

OSG58A3131A
D1

U5G
10M02SCE144C8G
POWER

U5F
10M02SCE144C8G
BANK8
IO/DEV_CLRn
IO/DEV_OE
IO/VREFB8N0
IO/CONFIG_SEL
nCONFIG
IO/CRC_ERROR
IO/nSTATUS
IO/CONF_DONE

10k R15
10k R16
10k R17
10k

U5D
10M02SCE144C8G
BANK5
IO/VREFB5N0

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File: ReferenceSignalGenerationCircuit.kicad_sch	
Title: Reference Signal Generation Circuit	
Size: A3	Date: 2026-07-12
KiCad E.D.A. 10.0.4	Id: 1/1

Rev: 1.0